Linux Fundamentals Final Revision Assessment Production Server Challenge

DevOps & Cloud Engineering

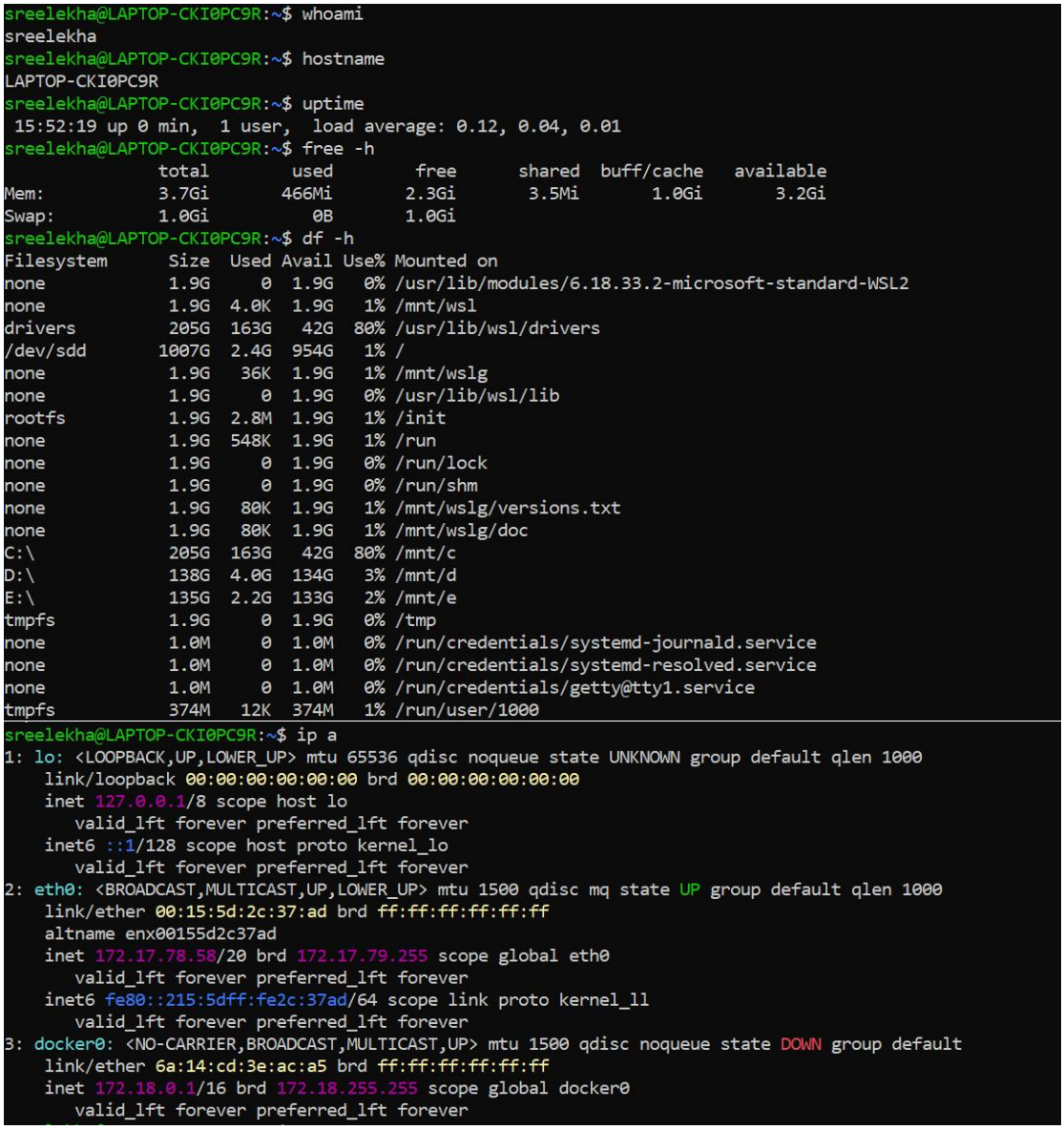
Final Linux Assessment covering Linux Basics, Filesystem, File Operations, Permissions, Process Management, Service Management, Storage, Backup and Troubleshooting.

Scenario Overview

You have joined a company as a Junior DevOps Engineer. A production Ubuntu server is reported to have multiple issues: Application inaccessible, disk space almost full, services not running,backup verification pending, permission issues reported, and management requires a health report.

Task 1 – Server Health Audit

Commands: whoami, hostname, uptime, df -h, free -h, top, ip a
Questions:
1. Current logged-in user?
Ans)whoami
2. Server hostname?
Ans)hostname
3. Server uptime?
Ans)uptime
4. Current RAM usage?
Ans)free -h
5. Current disk usage?
Ans)df -h
6. Current IP address?
Ans)ip a



Task 2 – Filesystem Investigation

Create company/app/logs, company/app/config, company/app/backups. Verify using tree.

Ans)mkdir -p company/app/logs company/app/config company/app/backups

- **mkdir — Creates new directories.**
- **-p — Creates parent directories if they do not exist.**
- **company/app/logs — Stores application log files.**

· company/app/config — Stores configuration files.

· company/app/backups — Stores backup files.

Verify Using tree:tree company



Questions: Absolute vs Relative path, mkdir -p, usefulness of tree.

Difference: An absolute path starts from the root directory (/), whereas a relative path depends on the current working directory.

The mkdir -p command creates directories along with any missing parent directories. It does not report an error if the target directory already exists

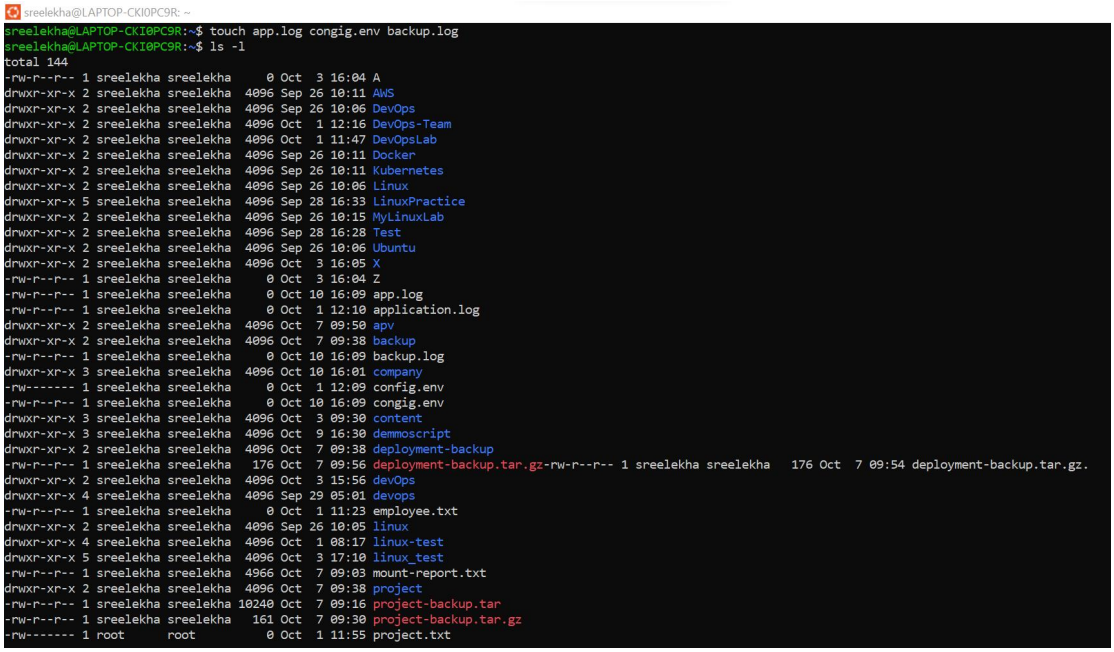
The tree command displays files and directories in a hierarchical, tree-like structure.(Example:tree company)

Usefulness:

- **Helps visualize directory structures.**
- **Verifies whether directories have been created correctly.**
- **Makes it easier to identify nested files and directories.**
- **Supports filesystem investigation and troubleshooting.**

Task 3 – File Operations

1) Create app.log, config.env, backup.log.



2) Perform cp, mv, rm operations.

1) cp command----------

cp app.log backup.log

{This copies the contents of app.log into backup.log. If backup.log already exists, its contents will be overwritten.}

cp app.log app-copy.log

{To copy a file while keeping the original unchanged, use a different destination filename}

2)Rename or move a file using mv----------

mv config.env app-config.env

{This renames config.env to app-config.env.}

To move a file into another directory:

mkdir -p company/app/config

mv app-config.env company/app/config/

3. Remove a file using rm----------

rm app-copy.log

{This deletes app-copy.log}

.

4. Verify the operations----------

ls -l
ls -l company/app/config/

{Check that the copied file exists, the renamed file is in its new location, and the removed file no longer appears.}

3)Questions: cp vs mv, rm vs rmdir, risks of rm -rf.

difference between cp and mv?

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cp (Copy): Copies a file or directory to another location while keeping the original unchanged.

mv (Move): Moves a file or directory to another location or renames it.

Difference: cp creates a copy, whereas mv moves or renames the original file.

What is the difference between rm and rmdir

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rm (Remove): Deletes files and can also delete directories when used with appropriate options.

Example:

rm app.log

rmdir (Remove Directory): Deletes empty directories only.

Example:

rmdir empty-folder

Difference: rm can delete files and, with options, non-empty directories. rmdir only removes empty directories.

risks of using rm -rf

The rm -rf command recursively deletes files and directories without prompting for confirmation in normal circumstances.

Example:

rm -rf folder-name

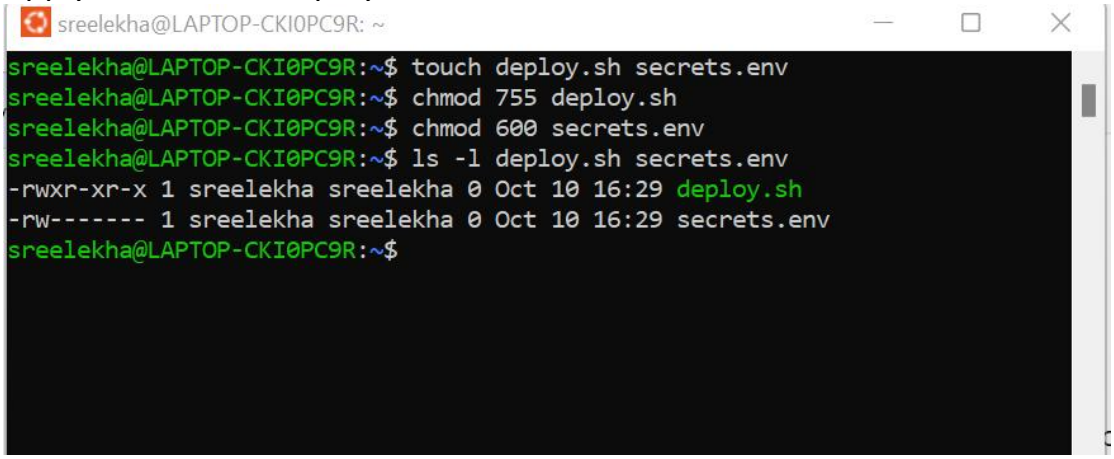
Risks:

- Permanently deletes files and directories without moving them to the Trash.
- Can delete important application files or system data.
- A wrong path can cause significant data loss.
- Wildcards or incorrect variables may target unintended files.
- Recovery may be difficult or impossible.

Task 4 – Users & Permissions

Create deploy.sh and secrets.env.

Apply chmod 755 deploy.sh and chmod 600 secrets.env.



Questions:

Why 755?

The chmod 755 deploy.sh command sets the file permissions so that the owner can read, write, and execute the file, while the group and others can read and execute it.

Why 600?

The chmod 600 secrets.env command allows only the file owner to read and write the file. The group and others receive no permissions.This helps protect sensitive configuration information, such as passwords, API keys, and tokens, from being read by other local users.

chmod vs chown?

chmod (Change Mode): Changes file or directory permissions.

Example: chmod 755 deploy.sh

chown (Change Owner): Changes the owner of a file or directory and can also change its group.

Example: sudo chown deployuser deploy.sh

To change both the owner and group:

sudo chown deployuser:deploygroup deploy.sh

Difference: chmod controls who can read, write, or execute a file. chown controls which user owns the file and, optionally, which group it belongs to.

Explain rwxr-xr-x.

The permission string rwxr-xr-x represents the permissions assigned to the owner, group, and others.

rwx — Owner can read, write, and execute

r-x — Group can read and execute but cannot write.

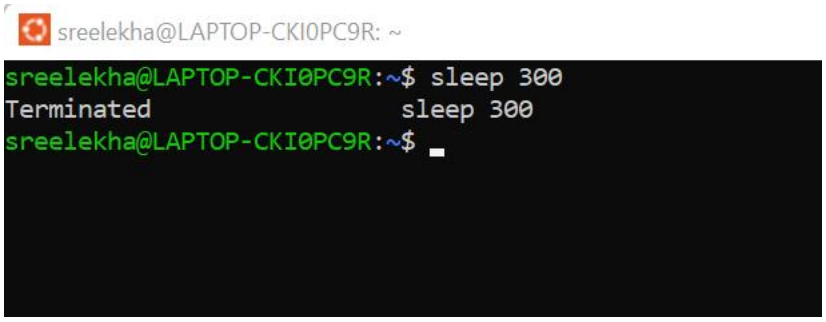
r-x — Others can read and execute but cannot write.

Numeric representation: 755

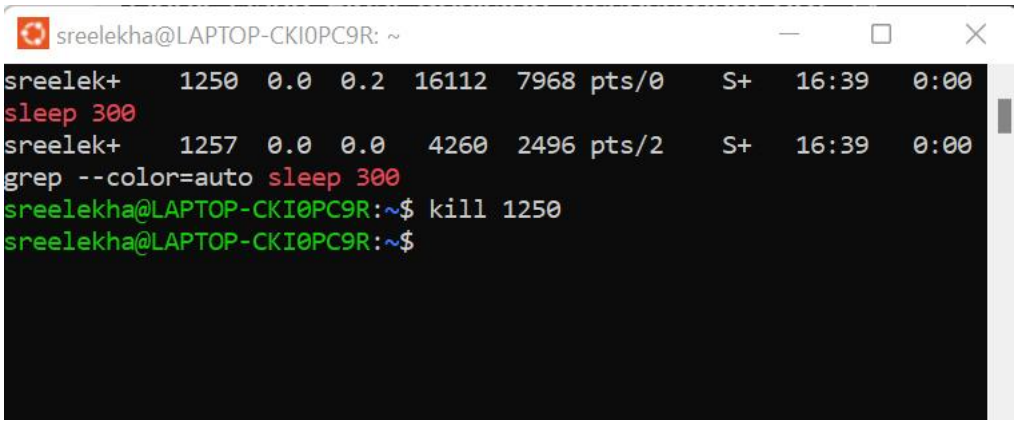
The leading - in the full listing -rwxr-xr-x indicates that the file is a regular file.

Task 5 – Process Troubleshooting

Create a background process using sleep 300 and Find PID and terminate it.



In another terminal



Questions:

PID:

PID (Process ID) is a unique number assigned to each running process in Linux

kill vs kill -9:

The kill command sends a signal to a process. By default, it sends SIGTERM (signal 15), which requests the process to terminate gracefully.
Kill -9 used to kill forcefully fot immidiate termination

htop usage:

htop is an interactive Linux process-monitoring tool that displays running processes and system resource usage in a user-friendly interface.

Main uses of htop:

Monitor CPU usage.

Monitor RAM and swap usage.

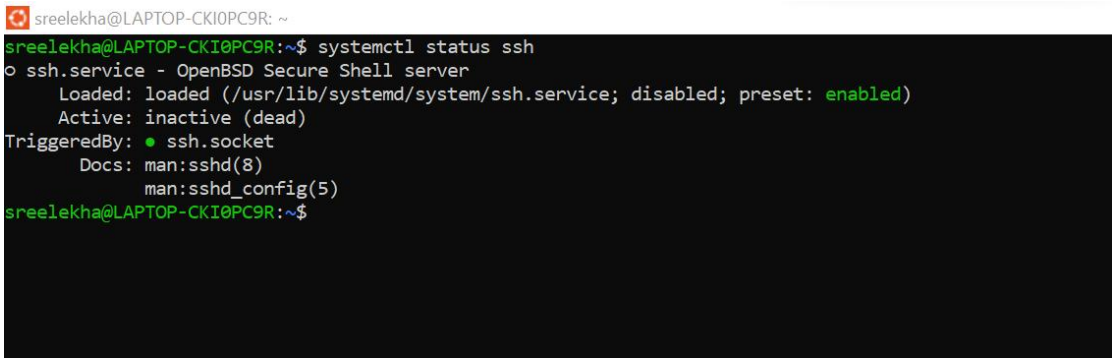
View running processes and their PIDs.

Sort processes by CPU or memory consumption.

Search for and filter processes

Task 6 – Service Investigation

Check SSH, Cron, loaded services and failed services.



```
sreelekha@LAPTOP-CKI0PC9R:~$ systemctl status cron
● cron.service - Regular background program processing daemon
     Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-10 15:51:38 UTC; 1h 1min ago
 Invocation: faac004a3b7f4d909489694e89f82dc3
       Docs: man:cron(8)
   Main PID: 120 (cron)
      Tasks: 1 (limit: 4472)
     Memory: 1.3M (peak: 3.3M)
        CPU: 40ms
     CGroup: /system.slice/cron.service
             └─120 /usr/sbin/cron -f -P

Oct 10 15:51:38 LAPTOP-CKI0PC9R systemd[1]: Started cron.service - Regular background program processing daemon.
Oct 10 15:51:38 LAPTOP-CKI0PC9R (cron)[120]: cron.service: Referenced but unset environment variable evaluates to an empty string: EXTRA_OPTS
Oct 10 15:51:38 LAPTOP-CKI0PC9R cron[120]: (CRON) INFO (pidfile fd = 3)
Oct 10 15:51:38 LAPTOP-CKI0PC9R cron[120]: (CRON) INFO (Running @reboot jobs)
Oct 10 16:17:01 LAPTOP-CKI0PC9R CRON[1096]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 10 16:17:01 LAPTOP-CKI0PC9R CRON[1098]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
Oct 10 16:17:01 LAPTOP-CKI0PC9R CRON[1096]: pam_unix(cron:session): session closed for user root
sreelekha@LAPTOP-CKI0PC9R:~$
```

```
sreelekha@LAPTOP-CKI0PC9R:~$ systemctl list-units --type=service
  UNIT                                     LOAD   ACTIVE     SUB          DESCRIPTION
  chrony.service                           loaded active     running      chrony, an NTP client/server
  console-setup.service                    loaded active     exited       Set console font and keymap
  containerd.service                       loaded active     running      containerd container runtime
  cron.service                             loaded active     running      Regular background program processing daemon
  dbus.service                             loaded active     running      D-Bus System Message Bus
  docker.service                           loaded active     running      Docker Application Container Engine
  getty@tty1.service                       loaded active     running      Getty on tty1
  keyboard-setup.service                   loaded active     exited       Set the console keyboard layout
  kmod-static-nodes.service                loaded active     exited       Create List of Static Device Nodes
  netplan-configure.service                loaded active     exited       Netplan Backend Configuration
  networkd-dispatcher.service              loaded active     running      Dispatcher daemon for systemd-networkd
  nginx.service                            loaded active     running      A high performance web server and a reverse proxy server
  rsyslog.service                          loaded active     running      System Logging Service
  setvtrgb.service                         loaded active     exited       Set console scheme
  snapd.seeded.service                     loaded active     exited       Wait until snapd is fully seeded
  systemd-binfmt.service                   loaded active     exited       Set Up Additional Binary Formats
  systemd-journal-flush.service            loaded active     exited       Flush Journal to Persistent Storage
  systemd-journald.service                 loaded active     running      Journal Service
  systemd-logind.service                   loaded active     running      User Login Management
  systemd-modules-load.service             loaded active     exited       Load Kernel Modules
  systemd-remount-fs.service               loaded active     exited       Remount Root and Kernel File Systems
  systemd-resolved.service                 loaded active     running      Network Name Resolution
  systemd-sysctl.service                   loaded active     exited       Apply Kernel Variables
  systemd-tmpfiles-setup-dev-early.service loaded active     exited       Create Static Device Nodes in /dev gracefully
  systemd-tmpfiles-setup-dev.service       loaded active     exited       Create Static Device Nodes in /dev
  systemd-tmpfiles-setup.service           loaded active     exited       Create System Files and Directories
  systemd-udev-load-credentials.service    loaded active     exited       Load udev Rules from Credentials
  systemd-udev-trigger.service             loaded active     exited       Coldplug All udev Devices
  systemd-udevd.service                    loaded active     running      Rule-based Manager for Device Events and Files
  systemd-user-sessions.service            loaded active     exited       Permit User Sessions
  ubuntu-fan.service                       loaded active     exited       Ubuntu FAN network setup
  unattended-upgrades.service              loaded active     running      Unattended Upgrades Shutdown
  user-runtime-dir@1000.service            loaded active     exited       User Runtime Directory /run/user/1000
  user@1000.service                        loaded active     running      User Manager for UID 1000
● wsl-pro.service                          loaded activating auto-restart Bridge to Ubuntu Pro agent on Windows
```

```
sreelekha@LAPTOP-CKI0PC9R:~$ systemctl --failed
  UNIT LOAD ACTIVE SUB DESCRIPTION

0 loaded units listed.
sreelekha@LAPTOP-CKI0PC9R:~$
```

Questions:

1. What is systemd?

systemd is a system and service manager used by many Linux distributions. It initializes the system during boot and manages services, processes, and other system units.

The systemctl command is used to interact with systemd.

Example:systemctl status ssh

Usefulness: Starts and stops services.

Checks service status.

Manages services during system startup.

Helps investigate service failures.

2. What is the difference between start, stop, restart, and reload?

- **Start:Starts a service that is not currently running.**

sudo systemctl start ssh

- **Stop:Stops a running service.**

sudo systemctl stop ssh

- **Restart: Stops and starts a service again. This is useful when a service needs a full restart to apply changes.**

sudo systemctl restart ssh

- **Reload:Asks a running service to reload its configuration without performing a full restart, if that service supports reloading.**

sudo systemctl reload service-name

Difference: start starts a service, stop stops it, restart starts it again, and reload refreshes its configuration without a full restart when supported.

3. Why should we check failed services?

Checking failed services helps administrators identify and troubleshoot problems that may affect system availability, performance, or functionality.

Reasons:

Detects services that failed to start.

Helps identify configuration errors.

Reveals missing dependencies or permission problems.

Helps investigate application outages.

Supports reliable server operation and maintenance.

Task 7 – Log Analysis

Use journalctl commands.

Journalctl-

This displays logs collected by the systemd journal. Press q to exit the log viewer.

Questions: Importance of logs, journalctl vs systemctl, crashed application investigation.

- ## Importance of logs in Linux?

Logs are records of events generated by the operating system, services, and applications.

Importance of logs:

- Help identify errors and system failures.
- Support troubleshooting and root cause analysis.
- Help monitor system activity and application behavior.
- Provide information about security-related events.
- Help administrators investigate performance issues.
- Support auditing and incident investigation.

Example:

journalctl -p err

This command displays error-priority messages available in the system journal.

Difference between journalctl and systemctl?

journalctl: Used to view and filter logs collected by the systemd journal.

Example:journalctl -u ssh -n 50

systemctl: Used to manage systemd services and inspect their status.

Example:systemctl status ssh

Difference: journalctl is used for log analysis, whereas systemctl is used for service management and status checks.

How do you investigate a crashed application?

A crashed application can be investigated by checking its service status, examining its logs, identifying errors, and verifying whether the problem has been resolved.

Step 1: Check the application service status

--->systemctl status app-service

Replace “app-service” with the actual service name.

Step 2: Review recent application logs

--->journalctl -u app-service -n 100

This displays the 100 most recent log entries for the service.

Step 4: Investigate the root cause

Check for:

- Configuration errors.
- Missing files or dependencies.
- Permission problems.

- Port conflicts.
- Memory or disk space exhaustion.
- Application exceptions or crashes.

Step 5: Fix the identified issue

Correct the configuration, permission, dependency, or resource problem identified during the investigation.

Step 6: Restart the service if appropriate

--->sudo systemctl restart app-service

Only restart after addressing the likely cause and confirming that restarting is appropriate.

Step 7: Verify the application

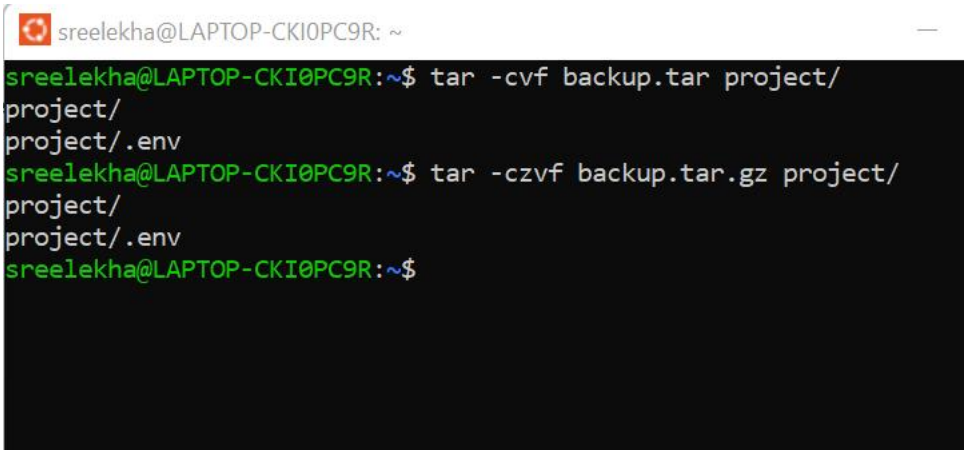
---->systemctl status app-service
---->journalctl -u app-service -n 50

Confirm that the service is running and that new logs do not show the same failure.

Conclusion: Logs help identify what happened, when it happened, and what may have caused the application failure. Investigating logs before restarting helps prevent repeated failures.

Task 8 – Storage & Backup

Create project backups using tar and compressed archives.



Questions: tar vs zip, backup importance, rsync in production.

Difference between tar and zip?

tar: Combines files into a single archive; compression is optional.

zip: Combines and compresses files into an archive.

Backups important?

- **Prevent data loss.**
- **Help recover files after system failures.**
- **Protect against accidental deletion.**
- **Support disaster recovery.**

rsync preferred in production?

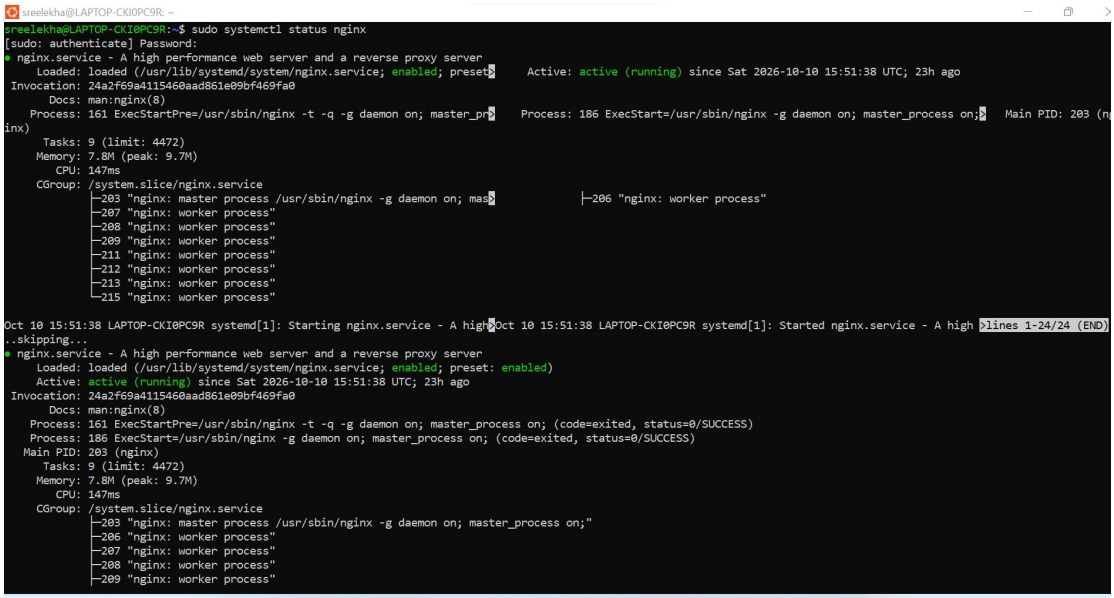
- **Transfers only necessary changes.**
- **Supports local and remote backups.**
- **Saves time and bandwidth.**
- **Preserves file attributes.**
- **Supports automated synchronization**

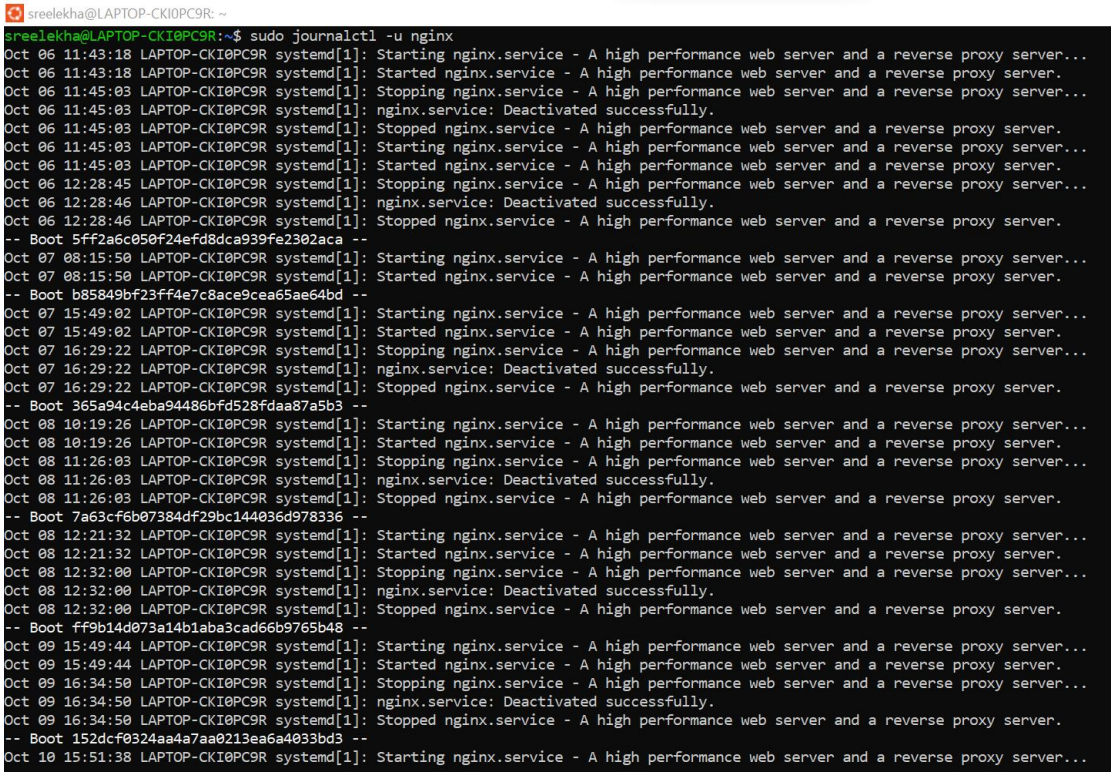
Task 9 – Website Down Scenario

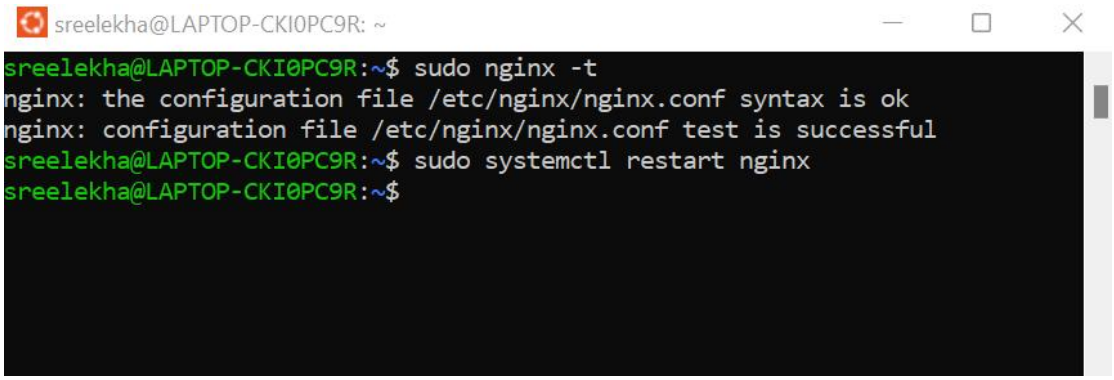
Investigate Nginx service issues using status, logs, config validation and restart procedures.

Ans)

- **sudo systemctl status nginx**
- **sudo journalctl -u nginx**
- **sudo nginx -t**
- **sudo systemctl restart nginx**







Questions:
What is Nginx? How to verify? What if restart fails?

1. What is Nginx?

Nginx is a web server used to host websites and handle HTTP requests.

2. How to verify Nginx is running?

Use sudo systemctl status nginx.

3. What if restart fails?

Check logs using journalctl, validate the configuration using nginx -t, fix errors, and restart Nginx.